

Smart Library Seat Management System

On-Demand Seating Allocation, Tracking, and Analytics Platform

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Project Proposal for UCS503P – Software Engineering Lab

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1 Higher-order Goal

Real-world impact

The Smart Library Seat Management System aims to transform the chaotic process of finding and retaining study spaces in college libraries into an organized, transparent, and fair digital experience. The system addresses the common issue of students blocking seats for extended periods while away, leading to wasted capacity and frustration for others.

The engineering objective is to build a robust software system that coordinates students, seating inventory, time-based policies, and real-time mapping through a unified workflow. This project offers a direct real-world use case while providing substantial software complexity to fulfill the learning outcomes of UCS503.

2 Problem Statement

In peak academic periods, library seating becomes a scarce resource. Existing methods rely entirely on physical roaming, which is time-consuming and inefficient.

The major problems are:

- **Uncertain availability:** Students waste time navigating multiple floors to find an empty seat.
- **Unfair Resource Blocking:** Students leave their belongings to 'reserve' a seat for hours while they are out for lunch or classes.
- **Poor Workflow Visibility:** Librarians cannot track which seats are genuinely occupied versus merely blocked.
- **No Data-Driven Planning:** Lack of analytics regarding peak library hours, preferred zones, and overall seating utilization limits effective infrastructure planning.

3 Proposed Solution

Software Proposition

The system will be developed as a responsive web-based platform connecting students and library administrators.

The proposed system will provide:

- **Real-Time Floor Map:** A visual representation of seat statuses (Available, Occupied, Temporarily Away) across different floors.
- **Smart Seat Finder:** Recommendations based on user preferences (e.g., quiet zone, charging port).
- **Temporary Away Permits:** An enforced countdown timer (e.g., 30 minutes) allowing students to take breaks. If they overstay, the seat is automatically marked as available.
- **Fair-Use Algorithms:** Limits on the number of breaks a student can take per day to prevent abuse.
- **Operations Dashboard:** Administrators can monitor live occupancy, override seat statuses, and view utilization metrics.

4 Core Workflow

The principal workflow is:

1. Student authenticates via their college ID/SSO.
2. Student checks the interactive floor map or uses the Smart Finder for recommendations.
3. Student selects an available seat and digitally checks in.
4. If a break is needed, the student activates a “Temporary Away” permit.
5. A countdown timer begins. The student must check back in before the timer expires.
6. If the timer expires, the seat is automatically released.
7. Upon leaving the library, the student checks out, permanently freeing the seat.

5 Why This is Suitable for UCS503

This project goes beyond a standard CRUD application. The core challenge involves coordinating real-time state management, time-based automated triggers, concurrent user access, and dynamic visual layouts.

This creates a practical software engineering problem involving:

- UML and dynamic state workflow modelling;
- Concurrency control and database lock design;
- API design for real-time state propagation (e.g., WebSockets);
- Agile backlog and sprint management;
- Software testing and state verification.

6 Mapping to Course Learning Outcomes

CLO	How it Demonstrates the CLO
CLO 1	An Agile/Incremental software process model will be adopted. The system can be delivered in discrete modules: authentication, static seat mapping, real-time status updates, and finally, the automated permit system.
CLO 2	Requirements elicited from students and librarians will be modeled using use-case, activity, sequence, and state diagrams, culminating in a formal SRS document detailing the strict policy rules for seat blocking.
CLO 3	The platform naturally supports a three-tier MVC architecture. It highlights the importance of low coupling (e.g., separating the countdown timer logic from the UI rendering logic) and cohesive API design.
CLO 4	The seat lifecycle (Available → Occupied → Away → Released) provides highly testable state transitions. Unit tests will validate timer expirations, while integration tests will ensure the UI updates correctly upon state changes.
CLO 5	The project will be managed using Agile methodologies, including user stories, sprint backlogs, and velocity tracking, heavily utilizing Git for version control and CI/CD pipelines.

7 Requirements Engineering

7.1 Stakeholders

- Students (End Users)
- Library Administrators (Operations)
- IT System Maintainers

7.2 Representative Functional Requirements

1. The system shall allow students to view real-time seat availability.
2. The system shall enforce a maximum time limit for the 'Temporary Away' status.
3. The system shall automatically transition a seat from 'Away' to 'Available' upon timer expiration.
4. The system shall prevent a single user from occupying multiple seats concurrently.
5. The system shall generate daily occupancy reports for administrators.

7.3 Representative Non-functional Requirements

- **Performance:** Status updates should reflect on all connected clients within 2 seconds.
- **Reliability:** The system must handle high concurrent check-ins during exam periods.
- **Energy efficiency:** The UI will utilize efficient state polling or WebSockets to minimize client-side battery drain.

8 UML and System Modelling

The project will produce the following models:

- **Use-case diagram:** Student check-ins vs Admin overrides.
- **Activity diagram:** The Temporary Away workflow and timer expiration.
- **Sequence diagram:** Client-server interaction during a concurrent seat booking attempt.
- **State diagram:** Seat lifecycle (Available, Occupied, Away, Maintenance).
- **ER diagram:** Relational data model for Users, Seats, Zones, and Sessions.

9 Software Architecture

A three-tier architecture with MVC principles is proposed.

- **Presentation layer:** Responsive web interface.
- **Application layer:** REST/WebSocket APIs handling timers and concurrency logic.
- **Data layer:** Relational database containing layout metadata and historical sessions.

10 Green Software Engineering

The project will incorporate:

- Event-driven WebSocket updates instead of aggressive client-side HTTP polling.
- Dark-mode support for OLED power savings.
- Minimal payload sizes for API responses to reduce network overhead.

11 Agile Development Plan

The project will be organized into short sprints.

Sprint	Deliverable
Sprint 1	Requirements, stakeholder analysis, SRS outline, and DB schema.
Sprint 2	Authentication and basic seat CRUD operations.
Sprint 3	Interactive Floor Map UI and real-time state propagation.
Sprint 4	The Temporary Away Timer engine and automated release logic.
Sprint 5	Admin Dashboard, analytics, and penalty enforcement logic.
Sprint 6	Testing, CI/CD pipeline finalization, and deployment.

12 Summary

The Smart Library Seat Management System solves a critical logistical issue in educational environments. Its strongest value for UCS503 is that it encompasses the complete software engineering lifecycle—from complex state modeling and concurrency handling to Agile development and Green Software practices.